

# BUSINESS ANALYTICS REPORT

## Customer Behaviour & Business Performance Analysis

*Customer Churn Analysis and Retention Opportunity Identification*

|                  |                                                         |
|------------------|---------------------------------------------------------|
| <b>Company</b>   | ABC Communications Ltd                                  |
| <b>Dataset</b>   | Telco Customer Churn                                    |
| <b>Programme</b> | AnalystLab Africa — Data Analytics Internship Programme |
| <b>Project</b>   | Week 1 — Business Analytics Case Study                  |

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## 1. Business Context and Analysis Objective

ABC Communications Ltd seeks to better understand the customer behaviours and characteristics associated with churn in order to strengthen its customer retention strategy.

The objective of this analysis is to identify the customer segments with the highest churn rates, examine the factors associated with churn, and translate these findings into targeted and actionable retention initiatives.

The analysis addresses the seven business questions defined in the assignment: understanding the customer base, identifying high-churn segments, assessing the role of contract type, tenure, services and payment methods, and recommending actions management should take.

## 2. Customer Base Overview

The dataset contains 7,043 customers, including:

- 1,869 churned customers
- 5,174 retained customers
- Churn Rate: 26.54%
- Retention Rate: 73.46%

Overall, approximately one in four customers has churned. The customer base is also highly concentrated in Month-to-month contracts, which account for approximately 55% of all customers. This becomes particularly important when combined with the churn patterns observed across contract types.

## 3. Data Analysis & Visual Evidence

The exploratory analysis combines customer segmentation, numerical distributions and statistical relationships to identify patterns associated with churn. The full visual evidence set includes three bar charts, two pie charts, two histograms, one box plot and one correlation heatmap. The three decision-critical charts below summarise the strongest segmentation findings.

### 3.1 Churn Rate by Contract Type

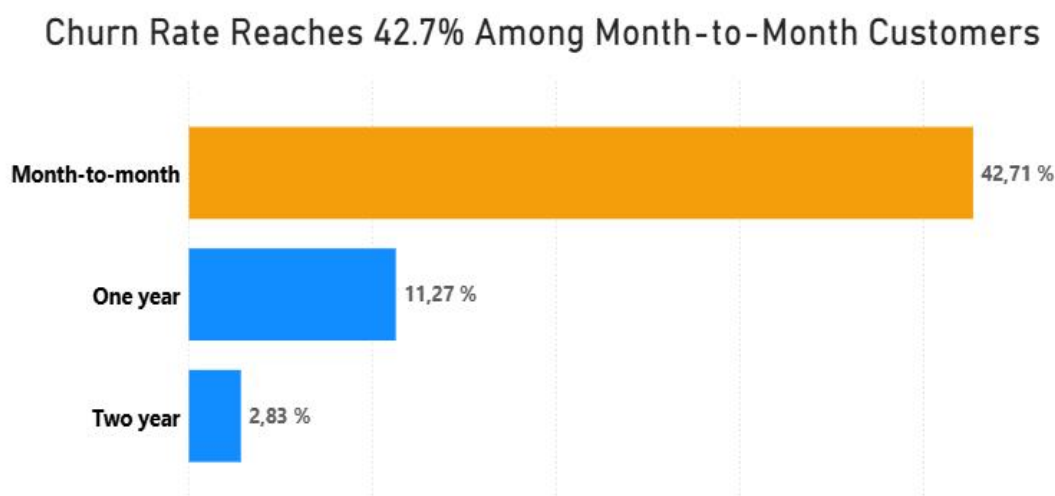


Figure 1. Churn Rate by Contract Type

Churn varies substantially across contract types: Month-to-month 42.71%, One year 11.27%, and Two year 2.83%. In addition, 88.6% of all churners are on Month-to-month contracts.

**Interpretation.** Month-to-month contracts represent one of the strongest observed churn risk markers. Customers on longer-term contracts are associated with substantially lower churn rates.

### 3.2 Churn Rate by Tenure Group

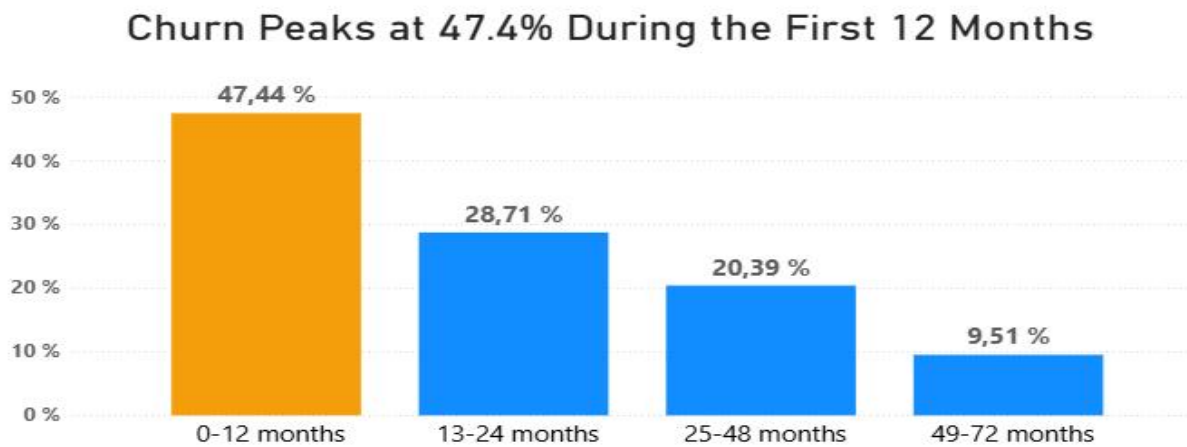


Figure 2. Churn Rate by Tenure Group

Churn declines progressively as customer tenure increases: 0–12 months 47.44%, 13–24 months 28.71%, 25–48 months 20.39%, and 49–72 months 9.51%.

**Interpretation.** The early stage of the customer relationship represents a critical retention window. Churn is highest during the first 12 months and decreases substantially as tenure increases.

### 3.3 Churn Rate by Internet Service

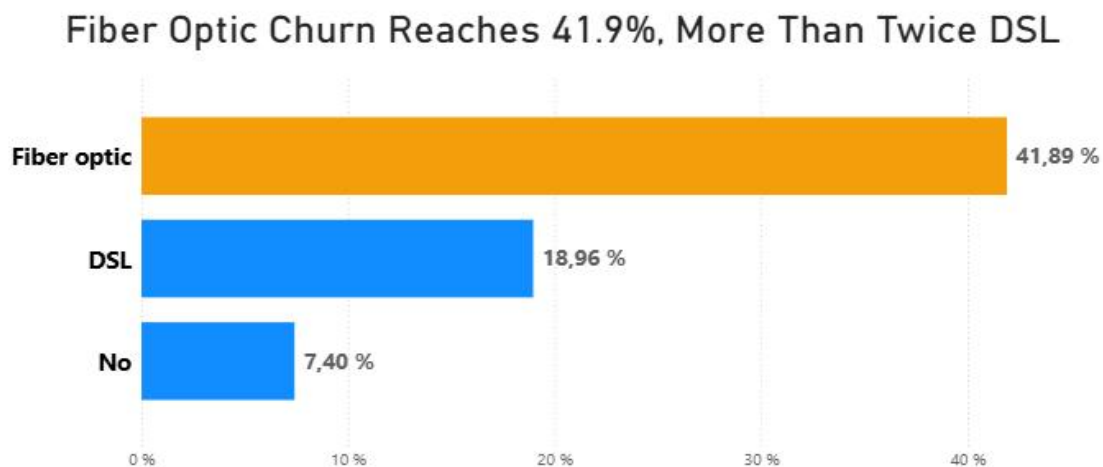


Figure 3. Churn Rate by Internet Service

Churn also varies substantially across Internet service categories: Fiber Optic 41.89%, DSL 18.96%, and No Internet Service 7.40%.

**Interpretation.** Fiber Optic customers exhibit more than twice the churn rate observed among DSL customers. Fiber Optic should therefore be treated as a priority segment for further diagnosis; this association alone does not demonstrate that the technology itself causes churn.

## 4. Consolidated Answers to the Business Questions

### 4.1 Which customer segments have the highest churn?

Churn is not evenly distributed across the customer base. Risk progressively increases as several characteristics are combined: Overall population 26.54% → Month-to-month 42.71% → + Fiber Optic 54.61% → + Electronic Check 60.37% → + Tenure 0–12 months 71.16%. This progression identifies a particularly exposed cumulative high-risk segment. These characteristics should be interpreted as associated risk markers rather than demonstrated causal factors.

### 4.2 Does contract type influence retention?

Churn differs substantially across contract types: 42.71% for Month-to-month customers compared with only 2.83% for Two-year customers. Contract type therefore represents a major churn risk marker within the analysed population.

### 4.3 Does tenure affect loyalty?

The relationship is particularly pronounced. Churn reaches 47.44% during the first 12 months, compared with only 9.51% among customers with 49–72 months of tenure. Customer retention risk is therefore heavily concentrated in the early customer lifecycle.

### 4.4 Which services are associated with churn?

Fiber Optic customers show the highest churn rate among Internet Service categories at 41.89%. A deeper analysis of support and security services within the Fiber Optic population shows: neither TechSupport nor OnlineSecurity 60.70%; TechSupport only 42.40%; OnlineSecurity only 39.71%; both 28.05%. The combined presence of TechSupport and OnlineSecurity is therefore associated with substantially lower churn within the Fiber Optic segment.

### 4.5 Which payment methods have higher churn?

Electronic Check emerges as an additional risk marker. When added to the Month-to-month + Fiber Optic profile, the observed churn rate increases from 54.61% to 60.37%. Electronic Check should therefore be incorporated into customer risk prioritisation, while avoiding the unsupported conclusion that the payment method itself causes churn.

## 5. Five Key Business Insights

**Insight 1 — Churn is heavily concentrated among Month-to-month customers.** Their churn rate reaches 42.71%, compared with 11.27% for One-year and 2.83% for Two-year contracts. They also account for 88.6% of all churners.

**Insight 2 — The first 12 months represent the main customer vulnerability window.** Churn reaches 47.44% during months 0–12 and progressively declines as tenure increases.

**Insight 3 — Fiber Optic customers represent a priority risk segment.** Their churn rate reaches 41.89%, more than twice the rate observed among DSL customers.

**Insight 4 — Churn risk intensifies when multiple risk markers accumulate.** Customers combining Month-to-month contracts, Fiber Optic, Electronic Check and 0–12 months of tenure reach an observed churn rate of 71.16%, compared with 26.54% for the overall customer population.

**Insight 5 — Lack of both TechSupport and OnlineSecurity is associated with substantial vulnerability among Fiber Optic customers.** Within this segment, churn reaches 60.70% when neither service is present, compared with 28.05% when both are present.

## 6. Three Business Risks

### Risk 1 — Early customer loss

The high churn observed during the first 12 months creates a risk of losing customers before sufficient customer lifetime value can be developed.

### Risk 2 — Churn concentration within a major commercial segment

Month-to-month customers represent approximately 55% of the customer base while accounting for a disproportionate share of churners. Further deterioration within this segment could materially affect customer retention performance.

### Risk 3 — Vulnerability within specific service configurations

Fiber Optic customers, particularly those without TechSupport and OnlineSecurity, exhibit high churn rates. This may signal issues related to customer experience, perceived value or service configuration that require further operational investigation.

## 7. Three Business Opportunities

### Opportunity 1 — Move from reactive to preventive retention

The cumulative risk profile enables management to identify priority customers before churn occurs, creating an opportunity for targeted preventive retention.

### Opportunity 2 — Strengthen onboarding and the first 12 months

The substantial decline in churn as tenure increases suggests that improving the early customer experience could represent a major retention opportunity.

### Opportunity 3 — Develop value through support and security services

Among Fiber Optic customers, the combined presence of TechSupport and OnlineSecurity is associated with substantially lower churn. This creates an opportunity to test bundling, service activation and customer-support interventions.

## 8. Actionable Business Recommendations

### R1 — Implement an Early-Life Retention Programme

Prioritise customers during their first 12 months through structured onboarding, proactive satisfaction checks and rapid intervention when customer friction is detected.

### R2 — Deploy cumulative risk-based customer targeting

Prioritise customers combining Month-to-month + Fiber Optic + Electronic Check + 0–12 months tenure. The observed 71.16% churn rate makes this population substantially more urgent than the overall customer base.

### **R3 — Test migration from Month-to-month to longer-term contracts**

Offer targeted incentives for eligible customers to move toward One-year or Two-year contracts. The intervention should be tested rather than assuming that longer contracts inherently cause higher retention.

### **R4 — Conduct a dedicated Fiber Optic customer experience diagnosis**

Investigate service experience, incidents, perceived quality, price-to-value perception, support, security and activation journey. The current analysis identifies the high-risk segment but does not establish its operational root cause.

### **R5 — Pilot a TechSupport + OnlineSecurity bundle for high-risk Fiber Optic customers**

The difference between 60.70% churn without either service and 28.05% with both services provides a strong basis for a controlled retention pilot. Compare outcomes between targeted customers and an appropriate comparison group before scaling.

### **R6 — Include Electronic Check in customer risk prioritisation**

Use Electronic Check as an additional risk signal, particularly when combined with the other characteristics identified. The recommended action is not to remove this payment method, but to use it for targeting and investigate potential payment-journey friction.

## **9. Management Conclusion**

The analysis demonstrates that customer churn at ABC Communications Ltd should not be treated as a uniform phenomenon. The overall churn rate of 26.54% masks substantial differences across customer segments. Customers who are recently acquired, on Month-to-month contracts, using Fiber Optic and paying through Electronic Check form a particularly exposed population, reaching an observed churn rate of 71.16% when these characteristics are combined.

Management should therefore move away from undifferentiated retention campaigns toward a targeted, early and measurable retention strategy.

**1. Intervene early in the customer lifecycle.**

**2. Prioritise customers accumulating multiple churn risk signals.**

**3. Test and measure retention interventions before scaling them.**